IE University · School of Science & Technology
Master in Business Analytics & Data Science (MBDS-PT) - Capstone Project

From the Last Century to the Next: Discovering Who Carrefour's Customers Really Are

A Behavioural Customer Segmentation Pipeline at Hypermarket Scale

Final Written Report

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Code repository: github.com/mabusharkh/carrefour-customer-segmentation

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Ethics Statement

I hereby certify that this report and the accompanying presentation is my own original work in its entirety, unless where indicated and referenced.

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1. Objective

Carrefour Spain serves close to eleven million loyalty-programme members, who account for between 70% and 80% of all tickets issued across its stores. The company generates over €11 billion in annual revenue. Despite this exceptional data asset, the retailer's working customer segmentation remains demographic: customers are grouped by age, gender, and a small number of inferred household signals (for example, identifying families with babies because they buy nappies). Carrefour Spain states that this is "segmentation of the last century" - it captures who a customer is on paper, but not what they actually buy, when, how often, or in what combinations.

The business problem is therefore one of missed relevance and missed revenue. A demographic label cannot distinguish a weekend organic-fresh shopper from a weekday promotion hunter who happen to share the same age and gender, yet these two customers require entirely different offers, communication timing, and retention mechanics. Treating them identically wastes campaign spend on customers who will not respond and fails to protect the high-value customers who drive the bulk of revenue.

The aim of this work is to replace demographic grouping with a behavioural segmentation learned directly from purchase history, and deliver a reproducible approach that Carrefour's Data Science team can industrialise and run as customer behaviour evolves. Concretely, the project sets out to:

- learn a vector representation of every customer that encodes purchasing behaviour ("what they buy, and how, when, and how often") rather than demographics
- partition the customer base into a set of behavioural tribes, each tribe being internally coherent and clearly distinct from the others
- profile and name each tribe in commercial language so category managers and CRM teams can act on it
- document the end-to-end pipeline so it can be re-executed and extended after the project ends.

Source of value. The segmentation creates value on three fronts identified with the sponsor: incremental revenue (relevant cross-sell and upsell to the right tribe at the right moment), cost savings (eliminating wasted contacts to customers with near-zero reaction probability), and customer satisfaction (offers that match genuine shopping patterns). The unit of analysis is the individual loyalty customer; the prediction target is not a single number but a discrete tribe assignment that triggers a differentiated commercial action.

Beyond segmentation itself, the project aims to answer several business-critical questions for Carrefour Spain: Which customers generate the highest long-term commercial value? Which customer groups are most responsive to promotional campaigns and loyalty interventions? Which behavioural patterns indicate opportunities for cross-category recommendations? And how can modern Machine Learning approaches uncover customer relationships that traditional demographic segmentation methods fail to detect? These questions guided the design of the segmentation pipeline and the evaluation of its commercial usefulness.

2. Data Sources

2.1 Raw data

Carrefour supplied a six-month extract of raw, ticket-level transaction data for a sample of almost one and a half million customers drawn from its loyalty base. The data is deliberately historical (two to three years old). After ingestion and de-duplication, the working dataset comprised:

Entity	Count	Notes
Ticket line records	191,017,715	One row per product per ticket (raw granularity)
Unique sales tickets	20,026,724	Distinct shopping trips
Unique customers	1,482,715	Anonymised loyalty identifiers
Distinct products (master)	893,944	Item master / catalogue
Products actually purchased	117,701	SKUs appearing in ≥1 ticket
Stores	4	Store 7, Store 2, Store 6, Store 11

Each ticket line records the customer identifier, the product, the timestamp, the store, the price paid, and a promotion flag. A product-category mapping was also provided, giving each SKU a free-text product name (e.g. “San Miguel can”) and a category/family label (e.g. “beer”). This category table is small relative to the catalogue and is used as an auxiliary signal, not as the primary feature.

2.2 Preparation and processing

Because the raw extract exceeds 190 million rows, the intention was to run preparation on a distributed processing setup such as Apache Spark; however, with no such environment available, all preparation was performed in DuckDB, an embedded analytical engine able to process the dataset out-of-core on a single workstation. DuckDB was configured with an 8 GB memory limit (to match the available hardware), eight threads, and temporary spill storage on a secondary drive, so that aggregations completed within local hardware constraints. The processing steps were:

1. Cleaning and Data-quality treatment: The dataset required little cleaning, and only two adjustments were applied during preparation. First, zero-amount lines were kept as promotional or free items, and negative-amount lines were netted against their matching purchase as returns where applicable, so total spend reflects net rather than gross expenditure.
2. Basket reconstruction - grouping ticket lines back into baskets (the multiset of products bought by one customer on one trip), which form the “sentences” for product embedding.
3. Behavioural feature engineering - aggregating each customer’s full history into 17 behavioural features (Section 3.2), all referenced to a fixed snapshot date of 30 June 2022 for recency calculations.
4. Store enrichment - extracting each customer’s store usage from the raw tickets to add store_id as an 18th feature for the sub-clustering stage (Section 3.5).

Exploratory analysis revealed a highly skewed, effectively bimodal customer base. Roughly half of customers shop frequently, with a median of 13 visits and high spend, while a large tail shop only once or twice in the six-month window. This small core generates a disproportionate share of visits, revenue and product diversity. Product purchases follow a classic retail long-

tail, a handful of staples appear in almost every basket while most SKUs are bought rarely (only 117,701 of 893,944 catalogue products were purchased at all). These characteristics justified density-based clustering and weighted customer representations capable of separating high-value loyal shoppers from sporadic or one-time buyers.

2.3 Feature engineering

To strengthen the product representation, the text for each product (its product name concatenated with the product-category label) was embedded as a text vector and combined with the behaviour-based Item2Vec vectors. We first trialled an open-source all-MiniLM sentence-transformer model to generate these embeddings, but it handled the Spanish product names poorly. We therefore adopted OpenAI's text-embedding-3-small model (1,536 dimensions), which is multilingual and supports Spanish, producing more reliable semantic vectors across the catalogue. These text vectors capture semantic similarity between products that rarely co-occur in baskets but are conceptually related, and they give a usable signal for long-tail products that appear too infrequently for purely behavioural embedding. No additional customer-level or third-party demographic data was used; the segmentation is built from purchasing behaviour alone, consistent with the project objective.

Additional behavioural features were constructed at the basket (ticket) level and then aggregated to the customer, so that signals such as average basket size, visit frequency, and the weekend and weekday shopping ratios reflect behaviour at the natural unit of a shopping trip rather than as undifferentiated customer totals.

3. Methodology

The entire methodology follows the four-phase data pipeline which is product embedding, customer mathematisation, dimensionality reduction, and customer clustering - with a specific, justified technical choice at each phase, followed by a profiling-and-naming stage and a sub-clustering refinement. Figure 1 shows the end-to-end flow.

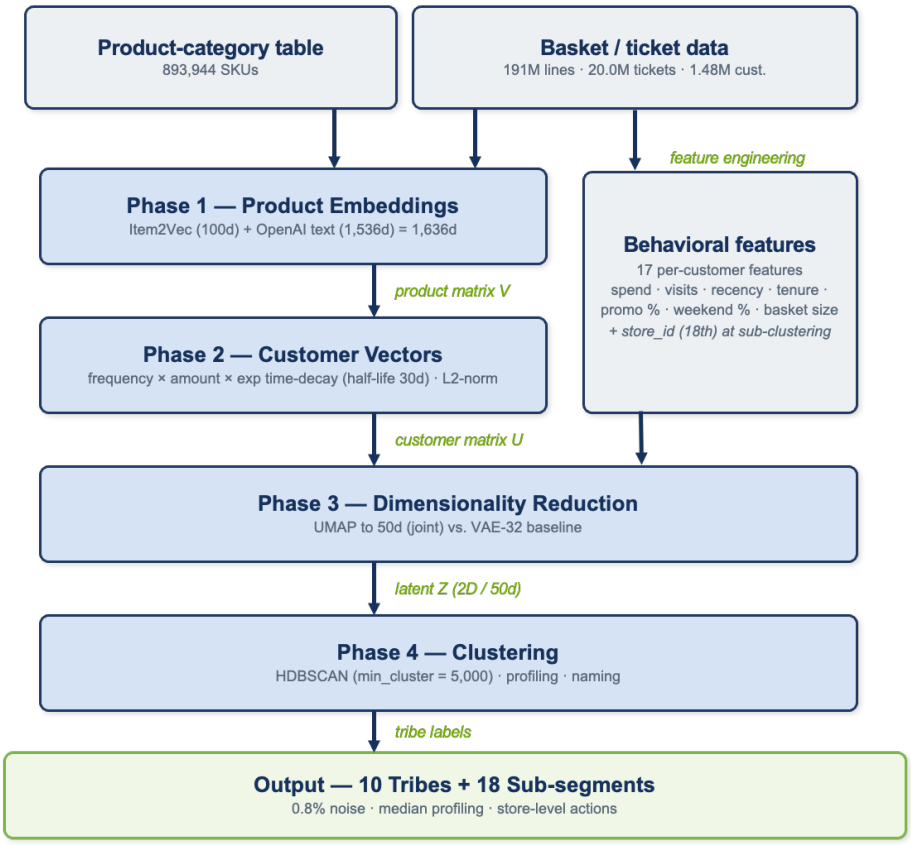


Figure 1. End-to-end segmentation pipeline, from raw baskets to named tribes and sub-segments.

3.1 Phase 1 - Product embeddings

Each basket is treated as a "sentence" of products. An Item2Vec skip-gram model (gensim, 100 dimensions) is trained over the basket corpus so that two products land close together when they are bought by behaviourally similar shoppers - the customer is the implicit context that defines co-occurrence. Because purely behavioural embeddings are unreliable for rare products (a SKU appearing in only a handful of baskets receives too few updates to learn a stable vector), the behavioural vectors are concatenated with the 1,536-dimensional OpenAI text embeddings of the product names, giving a 1,636-dimensional hybrid product matrix. The Item2Vec component is up-weighted by a tuned factor of 3.0× so that the behavioural signal dominates the text signal for products with sufficient purchase history, while the text component rescues the long tail. The methodological case for injecting the category table at training time (the Meta-Prod2Vec idea recommended in the sponsor brief) is discussed in Section 4 as the considered alternative.

3.2 Phase 2 - Customer mathematisation

A customer is expressed as a weighted combination of the product vectors they have actually purchased. A naive arithmetic mean is explicitly avoided because it gives a one-off impulse purchase the same weight as a weekly staple and treats a purchase from two years ago identically to one from last week. Instead, each purchase is weighted by two factors: how much the customer bought and spent, taken as the product of quantity and monetary amount (so a repeatedly bought, higher-value product counts more than a one-off impulse buy), and an exponential time-decay weight with a 30-day half-life ($\lambda = \ln 2 / 30$), so that recent purchases count more than old ones.

The per-purchase weight is therefore quantity × amount × exp(−λ·days_ago), where days_ago is counted from the fixed snapshot date of 30 June 2022; contributions are normalised to sum to one per customer, and each resulting customer vector is L2-normalised. This implements the sponsor's frequency-weighting and time-decay-curve recommendation directly.

3.3 Phase 3 - Dimensionality reduction

The 1,636-dimensional customer matrix is too high-dimensional for density-based clustering, and the consumer manifold is highly non-linear, so linear PCA is inadequate (it was retained only as a diagnostic baseline). Two non-linear reducers were trialled head-to-head - UMAP compressing the joint product-plus-behavioural representation to a 50-dimensional embedding - clustering is performed on these 50 dimensions, with a 2-D projection produced for visualisation only - and a variational autoencoder (VAE) compressing to a 32-dimensional latent space. Before compression the 17 behavioural features were scaled by $\sqrt{(1636/17)} \approx 9.81\times$ so they carry weight comparable to the 1,636-dimensional product signal rather than being swamped by it. The comparison and rationale for the chosen reducer are presented in Section 4; UMAP with joint compression was selected.

3.4 Phase 4 - Clustering, profiling, and naming

Customers are clustered with HDBSCAN, a density-based algorithm that discovers arbitrarily-shaped clusters of unequal size, handles variable density, and - critically - returns a genuine noise label (−1) for customers who do not belong to any tribe. It exposes a single, business-readable hyperparameter, min_cluster_size (set to 5,000 in the final run - the smallest tribe worth surfacing), with min_samples = 20. k-means was rejected because its spherical, equal-variance assumptions do not hold for behavioural tribes (Section 4).

Profiling and naming: Every cluster is automatically labelled by the label_cluster() function, which compares the cluster's value on each feature against the customer-base-wide reference and appends descriptive tags in a fixed order (Spend → Visits → Velocity → Tenure → Recency → Basket diversity → Private label → Weekend/Weekday → Promo). Skewed features (spend, visits, velocity, tenure, recency, unique products) are compared on cluster medians against the customer-base median, using 1.5× and 0.5× thresholds; bounded ratios (promo, weekend, private-label) are compared on cluster means against the cross-cluster mean. Using the median for skewed metrics is essential: cluster C0's mean spend is €49.44 versus a median of €17.35, so a mean-based rule would misclassify a one-time deal-hunter cohort as "High Spender". Per-tribe category lift and Item2Vec co-purchase pairs complete the commercial profile.

3.5 Sub-clustering with store identity

Macro-tribes, by design, average out internal variation. The four largest tribes (C7, C8, C9, C5 - together 94% of the base) were therefore re-clustered at a finer scale, adding store_id as an 18th behavioural feature. A customer's preferred store is treated as part of their shopping behaviour, not a demographic tag: it encodes format preference (hypermarket vs convenience), geography, and habitual channel, all of which correlate strongly with basket size and product mix. This produced 18 commercially actionable micro-segments which, together with the six tribes that were not sub-divided (C0, C1, C2, C3, C4 and C6), give 24 actionable leaf segments in total. store_id was deliberately held out of the main clustering: as a high-cardinality, near-categorical geographic variable it would otherwise fragment the macro-tribes along store and catchment lines and dominate the behavioural signal, so it is introduced only at the sub-clustering stage, where store-

format effects are the analytical target. The sub-clustering surfaced the single most impactful operational finding of the project (Section 6.2). A transparent caveat applies: because store choice partly reflects where a customer lives, some sub-cluster boundaries carry a geographic component - these are behavioural-geographic segments, which is commercially appropriate because Carrefour acts at store level.

4. Potential Solutions and Design Alternatives

At three points in the pipeline a genuine choice between competing methods existed. Each was evaluated and resolved on evidence; the most consequential - the dimensionality-reduction choice - was settled by running both candidates end-to-end on identical data in Sprint 9.

4.1 Dimensionality reduction: UMAP (joint) vs. Variational Autoencoder

Two reducers were run to completion and clustered identically with HDBSCAN. The results were decisive.

Criterion	UMAP + HDBSCAN	VAE + HDBSCAN	Winner
Behavioural features	Compressed jointly (9.81×)	Appended after (1.51×)	UMAP
Clusters discovered	10	6	UMAP
Noise rate	0.8% (~11,900)	1.7% (25,187)	UMAP
Geometry	Distinct blobs	Crescent / horseshoe	UMAP
Commercial granularity	High	Two blobs absorb nuance	UMAP

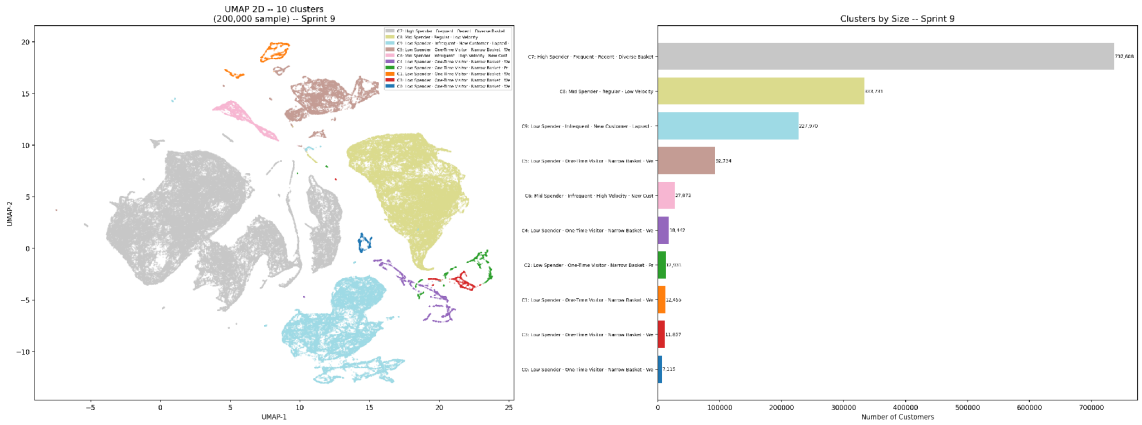


Figure 2. UMAP 2-D projection (200k sample) of the ten HDBSCAN tribes, with their size distribution. Clustering was performed on the 50-dimensional UMAP embedding; this 2-D view is for visualisation only.

UMAP outperformed the VAE on every criterion. The architectural reason is that UMAP compressed the behavioural features jointly with the product embeddings, so both signals informed the manifold simultaneously; in the VAE trial the encoder was trained on product embeddings only and behaviour was appended afterwards, leaving the latent space with no behavioural geometry. A 2-D projection of the VAE latent forms a crescent/horseshoe - the classic signature of a continuous gradient rather than discrete modes. HDBSCAN actually clustered on the 46-dimensional latent-plus-behavioural space, not on this 2-D view, but that continuous geometry is why it could only carve out coarse, density-threshold-dependent groups. The team tested the VAE both ways: appending the 14 behavioural features to the 32-dimensional latent (46-d) gave 6 coarse clusters at 1.7% noise, whereas compressing the behavioural signal jointly inside the latent - the

direct analogue of UMAP's approach - collapsed to just 2 clusters at 59.4% noise (Figure 3). Joint compression therefore succeeds for UMAP but fails for the VAE. By contrast, UMAP's distinct blobs indicate genuine separation. Decisively, UMAP separated a mid-frequency regular (C8) from a genuinely lapsed customer (C9), whereas the VAE collapsed both into one 635k-customer blob; those two groups require opposite interventions, so the distinction has direct commercial value.

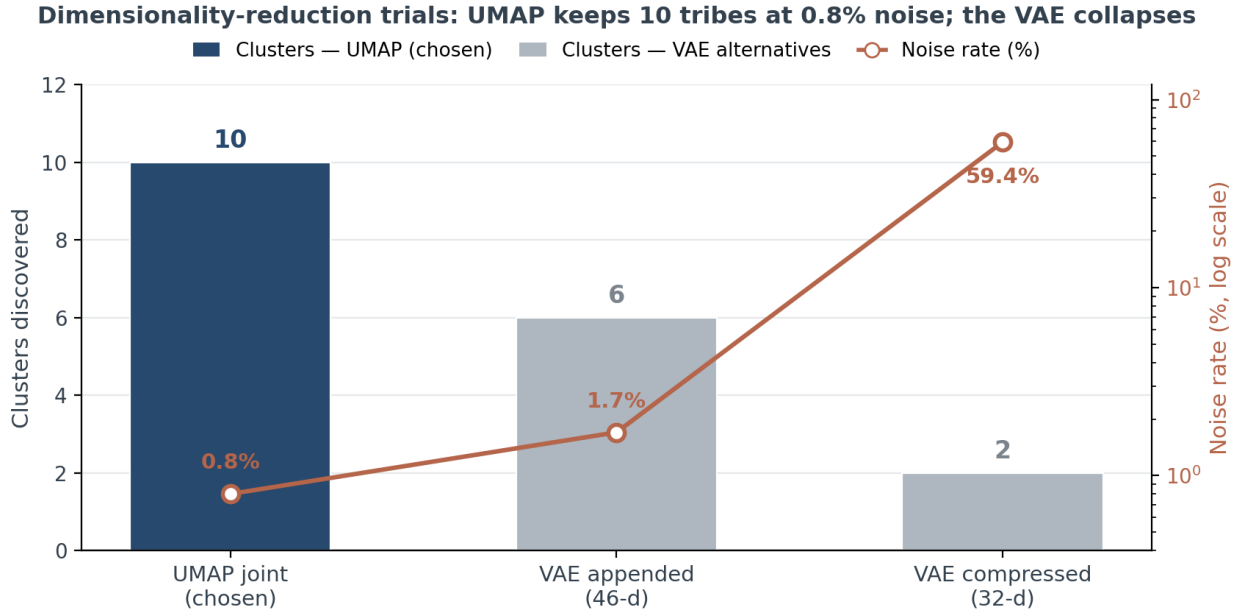


Figure 3. Dimensionality-reduction trials: UMAP yields 10 tribes at 0.8% noise, while the VAE collapses to 6 clusters (appended, 1.7% noise) or 2 clusters (jointly compressed, 59.4% noise).

4.2 Product embedding: hybrid Item2Vec + text vs. Meta-Prod2Vec

The adopted solution embeds products with behaviour-based Item2Vec concatenated with OpenAI text embeddings of product names. The principal alternative is Meta-Prod2Vec, which injects the product-category table into Item2Vec training as an auxiliary skip-gram loss, so that rare products inherit the well-learned behavioural signal of their category at training time rather than being patched afterwards. Meta-Prod2Vec is the more elegant long-tail solution and is the recommended path for a production v2. The hybrid text approach was chosen for this project because it was faster to implement reliably on the available stack, gave every product a vector regardless of purchase frequency, and let the team allocate scarce GPU time to the dimensionality-reduction comparison, which proved to be where cluster quality was actually won or lost. A pure post-hoc category-averaging back-off was rejected because it discards the basket signal for rare products entirely.

5. Development

The pipeline is implemented as a sequence of versioned Jupyter notebooks, one stage per notebook, so each phase can be re-run independently and audited. The full code is available in the project repository (GitHub Repository); the principal notebooks are listed below.

Notebook	Stage	Key output
product_embeddings_openai.ipynb	Phase 1	1,636-d product matrix
customer_vectors_duckdb_optimized_sprint9.ipynb	Phase 2	Customer matrix + 17 features
step4_umap_hdbscan_v3_sprint9.ipynb	Phases 3-4	UMAP 2D, tribes, profiles
step5/6_sprint9_*_subclustering_with_store.ipynb	Sub-clustering	18 micro-segments
tickets_eda.ipynb	EDA	Data-quality checks
step6_sprint9_product_dimension.ipynb	Product axis	40 product groups + cross-sell lift

Engineering practices: The development followed standard practices for clarity and reproducibility expected of field specialists: heavy data preparation pushed into DuckDB SQL for out-of-core performance; deterministic seeds set for Item2Vec sampling, the autoencoder, and UMAP; pinned library versions; and persisted artefacts (the product embedding table, the customer matrix, the UMAP model, and the HDBSCAN model) so that new customers can be scored without retraining the whole pipeline. The automatic label_cluster() routine keeps tribe naming consistent and comparable across sprints, removing subjective labelling.

Iterative delivery: The work was delivered over nine sprints. Early sprints established the embedding-to-clustering skeleton; the mid-project review delivered an eight-cluster segmentation; subsequent sprints corrected a series of feature-layer bugs (the weekend day-of-week encoding, average-hour, time-decay recency, negative amounts, and promotion-lift bias), tuned the VAE weight balance and the time-decay half-life, and re-validated profiling on medians. Sprint 9 - the final outcome reported here - added the UMAP-versus-VAE comparison and the store-aware sub-clustering, raising the segmentation from eight to ten tribes and lowering noise to 0.8%, the lowest of any trial.

Reproducibility and scale: The execution model deliberately avoids distributed training: a single workstation handles all four phases, with DuckDB providing the out-of-core data layer. This keeps the system debuggable and is exactly the property the sponsor needs in order to industrialise and re-run the pipeline internally. Finally, the segmentation, cluster profiles, co-purchase lifts and product-group drill-downs are surfaced through an interactive Streamlit application, giving Carrefour's commercial teams a point-and-click way to explore each tribe and sub-segment without opening the notebooks.

Business impact: Every design decision was made to produce a clear segmentation business teams could act on, not just a technically clean result. Clear separation with actionable segment sizes and narratives was treated as a deliverable in its own right, because segmentation is only useful to a category or CRM manager if it maps to a distinct, recognisable type of shopper.

Implementation notes: A few choices were made specifically to run at this scale on the available hardware. Customer aggregation and feature engineering run in chunked DuckDB passes that spill to disk, so the 191-million-line table is never held in memory. UMAP is fitted on a 150,000-customer sample and then applied to the full base in batches. The two non-numeric behavioural features (top category and peak hour) are dropped before scaling, and the remaining behavioural

block is scaled to balance it against the product vectors. HDBSCAN runs the Dual-Tree Boruvka algorithm over a KD-tree so that density clustering completes on a single machine. The final results are delivered through a Streamlit application that lets commercial teams explore every segment, compare them, and test campaign ideas without touching the underlying notebooks.

Technical Tools and Libraries

The following table lists every tool and library used in the project and its purpose.

Tool / library	Purpose in this project
DuckDB	Out-of-core SQL aggregation of 191M ticket lines on a single machine; data cleaning, returns netting, customer-level feature engineering.
Pandas	Smaller-scale transformations, merging outputs and preparing cluster profiles.
NumPy	Vector operations: concatenation, scaling, weighted averaging and matrix construction.
Gensim (Word2Vec)	Training the Item2Vec basket co-occurrence product embeddings (100-d).
OpenAI Embeddings API	Semantic product-name embeddings (text-embedding-3-small, 1,536-d) via external API.
scikit-learn	Feature scaling and nearest-neighbour reassignment of noise points.
umap-learn	UMAP dimensionality reduction of the customer and product spaces.
hdbscan / RAPIDS cuML	Density-based clustering; GPU (cuML) where available, CPU fallback otherwise.
PyTorch	Variational autoencoder (VAE) baseline trial for dimensionality reduction.
PyArrow / Parquet	Columnar storage of intermediate checkpoints (faster and smaller than CSV/JSON).
Matplotlib	Cluster scatter plots and diagnostic visualisations.
Streamlit	Interactive application for business users to explore tribes, sub-segments and product lifts.
Google Cloud (GPU instances)	Cloud GPU compute for the AE+HDBSCAN and Item2Vec/Prod2Vec trials, beyond the local workstation.
GPU (NVIDIA / cuML)	Accelerated UMAP, HDBSCAN and embedding training.
Claude Code	AI pair-programming assistance during pipeline development.
Claude Cowork	AI assistance for report drafting, data analysis and document preparation.

Serving and production. The segmentation, cluster profiles, co-purchase lifts and product-group drill-downs are delivered to business teams through an interactive Streamlit application (Annex G). The single-workstation setup used here is the development-scale equivalent of a production cloud architecture; the full local-to-cloud mapping, including a managed Spark or Databricks deployment for the eleven-million-customer scale, is set out in Annex H.

6. Conclusions

6.1 Technical result

The final pipeline segments 1,482,715 Carrefour Spain customers - derived from 191,017,715 ticket lines across 20,026,724 tickets - into ten behavioural tribes with a 0.8% noise rate, the lowest achieved across all trials. Algorithm choice was validated empirically: UMAP with joint compression beat a VAE baseline on cluster count (10 vs 6), noise (0.8% vs 1.7%), and geometric separation, and HDBSCAN was selected over k-means on the structural grounds that behavioural tribes are non-spherical and noisy. The ten tribes are summarised in the table below and Figure 4; sub-clustering the four largest tribes adds 18 micro-segments, for 24 actionable leaf segments in total. Three tribes account for 87.6% of the base.

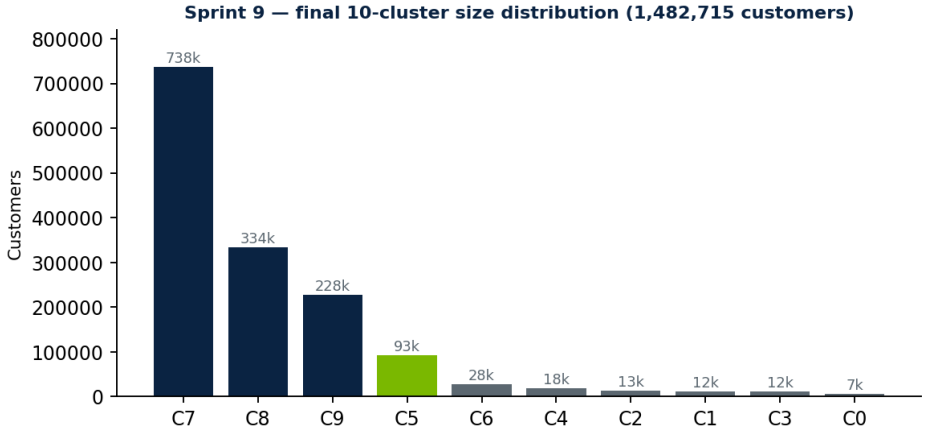


Figure 4. Final ten-tribe size distribution (Sprint 9, UMAP + HDBSCAN).

Tribe	Business Label	Customers	% customers	Behavioural label
C7	Revenue Engine	737,608	49.7%	High Spender · Frequent · Recent · Diverse Basket
C8	Loyal Regulars	333,731	22.5%	Mid Spender · Regular · Low Velocity
C9	Lapsed & One-Time	227,970	15.4%	Low Spender · Infrequent · New Customer · Lapsed · Narrow Basket · Weekday Shopper
C5	Weekend Fresh	92,734	6.3%	Low Spender · One-Time · Weekend
C6	Event Shoppers	27,872	1.9%	Mid Spender · Infrequent · High Velocity · New Customer · Lapsed · Narrow Basket · Weekend Shopper
C4	Convenience Grabs	18,442	1.2%	Low Spender · One-Time · Weekday
C2	Private Label Loyalists	12,931	0.9%	Low Spender · Private Label Loyalist
C1	One-Time Weekend	12,455	0.8%	Low Spender · One-Time · Weekend
C3	Cherry Pickers	11,857	0.8%	Low Spender · One-Time · Promo Hunter
C0	Cherry Pickers	7,115	0.5%	Low Spender · Extreme Promo Hunter (100%)

Sub-clustering the four largest tribes (C7, C8, C9 and C5) by store identity expands the ten tribes into 24 actionable leaf segments. The full set is listed below: the 18 store-aware sub-segments, together with the six smaller tribes that were left undivided and carry through unchanged.

Leaf segment	Customers	% of base	Behavioural label
C7 — High Spender · Frequent · Recent · Diverse Basket (737,608) — sub-segmented by store			
C7.0 Ultra-VIP	3,879	0.26%	Ultra-VIP volume buyer; very high spend and visits
C7.1 VIP multi-store	50,321	3.39%	VIP split across three stores; high value, lower per-store frequency
C7.2 VIP very frequent	229,836	15.50%	Very frequent high-value loyalist
C7.3 High-spender frequent	453,572	30.59%	Frequent high spender; some drifting recency (day-8 push target)
C8 — Mid Spender · Regular · Low Velocity (333,731) — sub-segmented by store			
C8.0 Frequent	17,275	1.17%	Frequent regular
C8.1 Regular	102,779	6.93%	Regular mid-spender (day-10 alert target)
C8.2 Weekend-concentrated	213,677	14.41%	Weekend-concentrated regular (Friday PM push target)
C9 — Low Spender · Infrequent · New/Lapsed · Narrow Basket · Weekday (227,970) — sub-segmented by store			
C9.0 Short-tenure	1,849	0.12%	Newly acquired, short tenure
C9.1 High-velocity promo	3,471	0.23%	High-velocity promotion-driven (PL substitution target)
C9.2 Regular beer buyer	1,445	0.10%	Niche regular beer buyer
C9.3 Infrequent regular	23,271	1.57%	Infrequent but repeat (frequency-nudge target)
C9.4 One-time / lapsed	197,934	13.35%	One-time or lapsed; triage with passive promo only
C5 — Low Spender · One-Time · Weekend (92,734) — sub-segmented by store			
C5.0 Weekend FMCG	4,070	0.27%	Weekend FMCG basket
C5.1 Electronics deal hunter	3,819	0.26%	Weekend electronics deal hunter (Friday electronics push)
C5.2 PL loyalist	5,628	0.38%	Private-label loyalist
C5.3 Weekend regular	9,352	0.63%	Weekend regular; highest sub-cluster value (retention reward)
C5.4 Weekend 2-visit	13,581	0.92%	Two-visit weekend shopper (visit-3 nudge)
C5.5 Weekend fresh mass	56,284	3.80%	Weekend fresh-counter mass shopper
Tribes left undivided (carried through unchanged as leaf segments)			
C6	27,872	1.88%	Mid Spender · Infrequent · High Velocity · New/Lapsed · Narrow Basket · Weekend
C4	18,442	1.24%	Low Spender · One-Time · Weekday
C2	12,931	0.87%	Low Spender · Private Label Loyalist
C1	12,455	0.84%	Low Spender · One-Time · Weekend
C3	11,857	0.80%	Low Spender · One-Time · Promo Hunter
C0	7,115	0.48%	Low Spender · Extreme Promo Hunter (100%)

Table 2. The 24 actionable leaf segments: 18 store-aware sub-segments of the four largest tribes plus the six undivided tribes. C8 uses the store-aware sub-cluster split, consistent with C7, C9 and C5 (see Annex A and Annex F).

6.2 Headline commercial findings

Finding 1 - A Store-Level Revenue Gap No Demographic System Would Ever Find. Sub-clustering with store_id revealed that the hypermarket (Store 7) systematically generates a 1.6-2.1× higher average ticket-line value than the lower-depth formats for identical behavioural profiles, a pattern that repeats across all five sub-clustered profiles (Figure 5). Store 7 is the consistent outlier: Store 2, Store 6 and Store 11 all cluster in the €2.13–€2.79 ticket-line range, while Store 7 delivers €3.89–€4.95. The behavioural-profile match rules out a crude customer-mix explanation - it is not simply that different types of shopper frequent the two formats. Two explanations remain to be separated. The first is an assortment effect: Store 7's greater product depth - notably the FMCG multipacks and full fresh-counter range absent from the smaller formats - lets a shopper build a higher-value basket. The second is a shopping-mission effect: a category-mix difference, such as the weekly hypermarket shop versus a quick top-up run, could lift per-line value on its own, independently of assortment. The profile match does not distinguish these two, because the embedding does not encode a shopping mission. To isolate the store effect we recommend two confirmations: comparing ticket-line value within product category, and analysing multi-store customers (the same individuals across formats) to remove the residual customer effect. If the gap holds under both, the lever is an SKU and assortment audit at the lower-depth stores - the highest-ROI intervention identified in this analysis. Whether the Store 7 model can be replicated is addressed in Section 6.4 (store-network format decision).

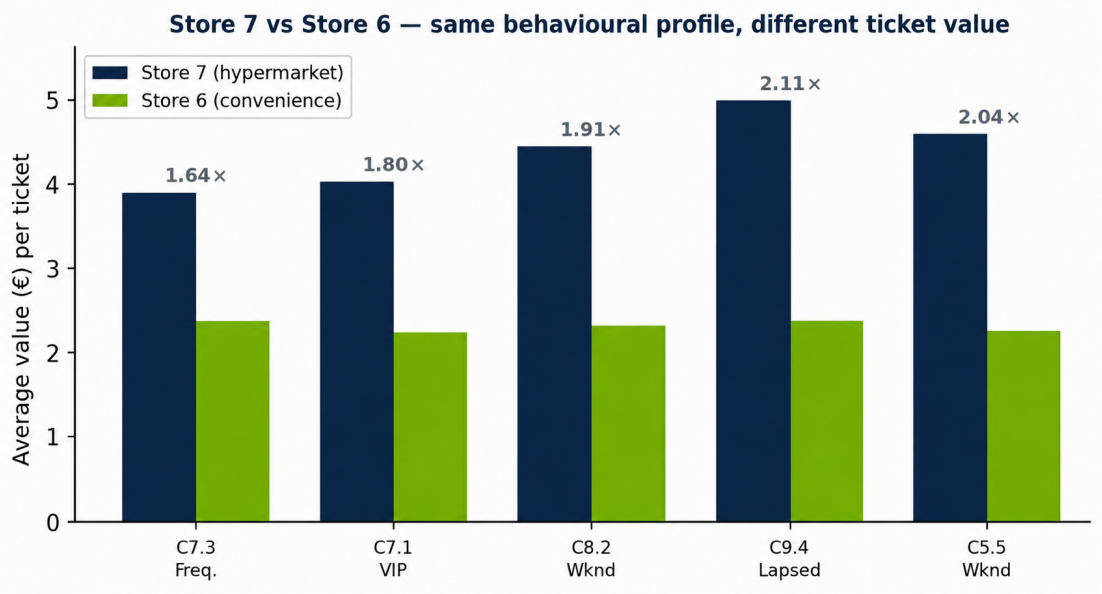


Figure 5. The same behavioural profile produces a 1.6-2.1× higher ticket-line value at Store 7 than at Store 6.

Finding 2 - your most valuable 3,879 customers are invisible.

C7.0: 3,879 customers, €1,055 median spend, 26 visits, 168-day tenure. Top products: Mahou 50cl, Red Bull, Voll-Damm - likely large families. Under the current system, they are indistinguishable from any other shopper. Under this segmentation, they are identifiable by name and worth a ‘Carrefour Pro’ tier.

Finding 3 - The Fresh Counter Is Your Most Powerful Retention Tool. Strawberries and bananas are the highest-penetration SKUs across every major tribe (325k strawberry buyers in C7 alone). They drive weekly visit cadence, so fresh-counter quality at Store 7 should be managed as a

retention KPI, not a merchandising metric: its degradation would reduce visit frequency for the 685k customers in C7.2 and C7.3 faster than any other variable.

Finding 4 - know where not to overinvest. C9.4: 197,934 one-time visitors, 59-day recency. A broad win-back wastes budget on 198,000 near-zero-probability customers. The right action: passive app push only, at zero marginal cost. Budget saved goes to C7.3 - 453,000 drifting-loyal customers who need a day-8 fresh counter offer before the gap becomes permanent.

A structural weekday/weekend divide underpins these findings. Seven tribes are 67–100% weekday shoppers; three are 100% weekend. This is a behavioural split, not a demographic one - in effect Carrefour serves two customer bases, with different product drivers and retention mechanics, through the same stores.

6.3 Business impact and practicality

The segmentation translates directly into sequenced, cost-aware actions. Immediate priorities are a Store 6 SKU audit (beer multipacks, fresh display, full private-label range) to close the ticket-value gap; a day-8 recency push for the 453k drifting-loyal C7.3 customers; weekend re-engagement for the highest-churn-risk C8.2; and a “Pro” outreach to the 3,879 ultra-VIPs. Medium-term actions include a cross-store loyalty mechanic for C7.1 and frequency nudges for recoverable infrequent-regular sub-segments. Crucially, the analysis also identifies where not to spend: broad win-back across the 198k one-time/lapsed C9.4 customers would yield negligible ROI and should be replaced by passive, promotion-triggered channels - a direct cost saving. Because the pipeline runs reproducibly on a single workstation and persists all scoring artefacts, Carrefour’s team can industrialise it and re-segment on a quarterly cadence, which is the sponsor’s stated end goal.

6.4 Business actions by cluster

The segmentation maps directly onto differentiated commercial actions. The table below consolidates the recommended action for each tribe; store-level execution detail and the sub-segment playbook sit in the sub-cluster analysis.

Five strategic decisions. These high-level decisions precede any segment-specific action; they are derived from patterns that repeat across multiple clusters and sub-clusters.

Decision	Rationale	Segments	Priority
1. Mitigate the C7 concentration risk	49.7% of the customer base sits in one behavioural profile, so any operational failure at Store 7 hits half the revenue base at once.	C7.0 to C7.3	Immediate
2. Automate the C8 retention lifecycle	333,731 mid-spenders show recency drift. Left unmanaged, C8 migrates to C9; a day-10 automated intervention reaches 22% of the customer base.	C8	Immediate
3. Build segment-aware promotions	Deep promotions on olive oil and beer multipacks generate C0 and C3 cherry-picker footfall, not C7/C8 loyalty. The promotional calendar must filter through segment identity.	C0, C3, C9.1	Q3 2026
4. Rebuild loyalty around the brand	C7.1 (50,321 VIPs) split evenly across three stores. A store-based programme counts them as three low-frequency shoppers; a brand-level mechanic unifies them.	C7.1, C5.0	Q3 2026
5. Resolve the store-network format question	Store 7 generates 1.6 to 2.1 times higher ticket-line value than every other store. This is a capital-allocation and format-strategy decision, not an SKU fix.	All segments	Q4 2026

Tribe	% Customers	Recommended business action
C7	49.7%	Protect the revenue engine. Manage fresh-counter availability at Store 7 as a retention SLA; send a day-8 recency push to the drifting C7.3 sub-tribe; create a Carrefour Pro tier for the 3,879 ultra-VIP volume buyers (C7.0).
C8	22.5%	Retain before drift. Automate a day-10 to day-14 fresh-counter offer to arrest recency drift before C8 lapses into C9. Highest-volume lifecycle decision.
C9	15.4%	Triage, do not blanket-spend. Passive, promotion-triggered app push only for the 197,934 one-time C9.4 customers; targeted frequency nudges for the recoverable C9.3 regulars.
C5	6.3%	Activate weekends. Fresh-counter bundles to convert one-time weekend visitors; visit-3 loyalty reward for the C5.3 weekend regulars.
C6	1.9%	Convert event shoppers. Second-visit offer within 14 days for these high-velocity, weekend, event-driven buyers.
C2	0.9%	Reinforce own-brand loyalty. Private-label stamp card and adjacent own-brand cross-sell for the 100% private-label loyalists.
C0, C3	1.3%	Filter from broad promotions. One-time promo cherry-pickers (C0 at 100% promo, one-day tenure); exclude from deep-discount campaigns aimed at loyal tribes.
C1, C4	2.0%	Low-cost reactivation only. Minimal passive outreach for one-time weekend (C1) and weekday convenience (C4) visitors.

Store overlay (highest-ROI immediate action): audit and stock Store 6 to close the 1.6x to 2.1x ticket-value gap versus Store 7, covering beer multipacks (Mahou 12x33cl and 28x33cl), the fresh strawberry and banana display, and the full Carrefour private-label range.

Implementation timeline. Actions are sequenced by commercial impact and deployment cost: Phase 1 is cheapest with the highest reach; Phase 4 requires structural decisions and longer planning horizons.

Phase	Timeline	Actions	Reach
Phase 1	0 to 1 month	Store 7 fresh-counter SLA; C7.3 day-8 push; C8.2 Friday PM push; Store 6 SKU audit; C7.0 direct outreach.	~670K customers
Phase 2	1 to 3 months	C8.1 day-10 alert; C9.3 frequency nudge; C5.4 visit-3 nudge; C5.1 Friday electronics push; C2 private-label stamp card.	~152K customers
Phase 3	3 to 6 months	Cross-store loyalty mechanic; C5.3 retention reward; C3 PL beer conversion; C9.1 PL substitution; Store 2 product standardisation.	~75K customers
Phase 4	6 to 12 months	Segment-aware promo calendar; store-network format decision; Carrefour Pro tier launch; pipeline re-run on fresh data.	Full customer base

6.5 Limitations and future work

- **Architectural limitations.** Using the model on a longer timespan, and on a wider set of clients (from 1.5 million to 11+ million members in El Club Carrefour) would be challenging on the developed architecture based primarily on Pandas and DuckDB. Adopting distributed big data computing might be necessary. Tools such as Apache Spark could be helpful, however vertical scaling to a system with much stronger RAM memory and with multiple GPUs for dimensionality reduction and clustering could prove to be enough.
- **Geographic confounding.** Using store_id as a behavioural feature introduces residential geography into some sub-cluster boundaries; this is commercially appropriate but should be stated in executive communication.
- **Static snapshot.** Tribes are trained on a historical six-month window; production use requires a quarterly (or streaming) re-segmentation cycle as behaviour evolves.
- **Loyalty programme only.** The provided dataset includes a limited set of customers who most likely can be identified by multiple purchases through an ID number. They are members of El Club Carrefour. To understand and serve customers who are not members of the club - adding different segmentation approaches would be necessary.

Future directions include adopting Meta-Prod2Vec for stronger long-tail embedding, testing the addition of demographic features for stability, building a real-time scoring API that assigns new customers to the nearest tribe on first purchase, and running controlled A/B tests on the store-level recommendations to measure causal lift rather than correlational difference.

Annex A - Tribe and Sub-Segment Profiles

Per-tribe profile values. Spend, visits, tenure, days-since-purchase and unique products are reported as cluster medians (robust to the heavy right-skew of retail data); the ratio columns - Weekend % and Promo % - are cluster means, consistent with the labelling rules in Annex B.

Tribe	Med spend	Med visits	Tenure	Days since	Uniq prod.	Weekend %	Promo %
C7	€465	13	156 d	8	100	28%	22%
C8	€162	7	131 d	17	39	33%	20%
C9	€43	2	2 d	56	11	0%	20%
C5	€31	1	1 d	75	9	100%	20%
C6	€99	2	14 d	54	24	100%	23%
C2	€5	1	1 d	-	-	low	0%
C0	€27	1	1 d	-	1	0%	100%

Sector co-purchase lift (C7, the core base). Every sector pair exceeds 1.0×, confirming genuine whole-store cross-shopping: FMCG×Fresh 1.12×, Home×FMCG 1.20×, FMCG×Clothing 1.46×, Electronics×Clothing 1.68×. For C8 and C9 nearly all pairs fall below 1.0×, indicating deliberate single-category shopping.

Selected Sub-Segments (Store-Aware)

Sub-segment	Customers	Spend	Visits	Defining characteristic
C7.0 Ultra-VIP	3,879	€1,055	26	Beer-volume buyers; likely large families; 73% Store 7
C7.1 Multi-store VIP	50,321	€659	25	Split evenly across 3 stores; brand- not store-loyal
C7.3 Drifting loyal	453,572	€418	11	Widest recency in C7 (11 d) - earliest drift signal
C9.4 One-time/lapsed	197,934	€40	1	86.8% of C9; near-zero reactivation - passive only
C9.3 Infrequent regular	23,271	€71	4	Recoverable; 21-day cadence; frequency-nudge target
C5.1 Electronics hunter	3,819	€22	1	95% promo; weekend; TVs/Switch; 87% Store 7
C5.2 PL loyalist	5,628	€4.5	1	89% private label - highest in customer base

Store-format gap (average ticket-line value €; Store 7 / Store 2 / Store 6). C7.3 €3.89 / €2.57 / €2.37 (1.64×); C7.1 €4.01 / €2.52 / €2.23 (1.80×); C8.2 €4.40 / n/a / €2.30 (1.91×); C9.4 €4.95 / €2.79 / €2.35 (2.11×); C5.5 €4.55 / €2.65 / €2.23 (2.04×) - ratios are Store 7 vs Store 6. Store 2, Store 6 and Store 11 all cluster in the €2.13–€2.79 range.

Annex B - Cluster Labelling Rules

Tags are appended in fixed order. Skewed metrics use cluster median vs cross-cluster median; bounded ratios use cluster mean vs cross-cluster mean. One-Time Visitor (visits ≤ 1) suppresses the velocity, tenure, and recency tags, which are undefined for single-visit customers.

Feature	High tag	Low tag	Threshold
total_spend	High Spender	Low Spender	1.5× / 0.5× median
total_visits	Frequent	Infrequent	1.5× / 0.5× median
spending_velocity	High Velocity	Low Velocity	1.5× / 0.5× (tenure ≥14 d)
tenure_days	Long Tenure	New Customer	1.3× / 0.5× median
days_since_last	Lapsed	Recent	1.5× / 0.5× median
unique_products	Diverse Basket	Narrow Basket	1.3× / 0.7× median
private_label_ratio	PL Loyalist	-	> 1.4× mean
weekend_ratio	Weekend Shopper	Weekday Shopper	> 0.45 / < 0.20 mean
promo_ratio	Promo Hunter	-	> 1.5× mean

Annex C - Sprint Evolution

The segmentation was reached over nine iterative sprints, each introducing a single change so its effect on cluster count and noise could be isolated. The table below traces that progression, mirroring the mid-project review's visual comparison and extending it to the final Sprint 9 outcome reported in this document. Two autoencoder trials (Sprints 3 and 4 AE) were run and discarded after they consistently over-segmented the space.

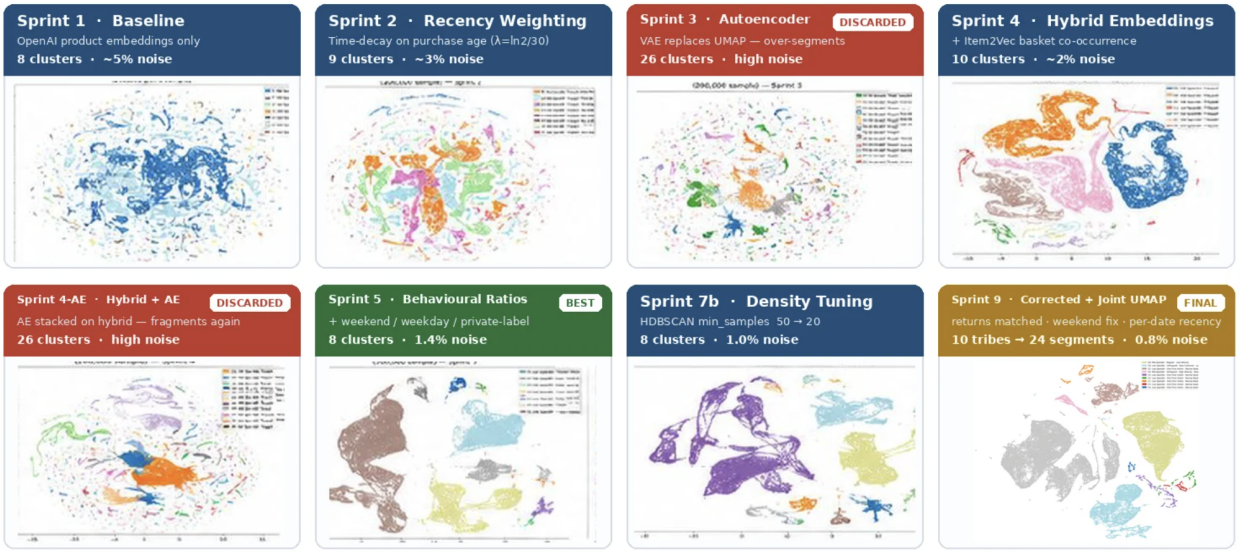


Figure E.1. Sprint evolution from Sprint 1 to Sprint 9: clusters discovered and noise rate at each iteration, with the two discarded autoencoder trials (Sprint 3 and Sprint 4-AE) marked.

Sprint	Change introduced	Clusters	Noise	Outcome / status
1 - Baseline	OpenAI text embeddings only; no time-decay or basket features.	8	~5%	Two dominant clusters absorbed most customers; the remainder fragmented into ribbon-like micro-clusters.
2 - Time-Decay	Added exponential time-decay weighting so vectors reflect recent behaviour.	9	~3%	Four clearer blobs emerged; interpretability improved markedly.
3 - Autoencoder (VAE)	Replaced UMAP with a Variational Autoencoder.	26	High	Over-segmentation into many redundant groups - Discarded.
4 - Hybrid Embeddings	Added Item2Vec basket co-occurrence vectors to the OpenAI embeddings.	10	~2%	Cleaner separation; main archetypes stable, validating the hybrid approach.
4 AE - Hybrid + AE	Re-applied the autoencoder on top of the hybrid embeddings.	26	High	Same fragmentation recurred, confirming AE was incompatible with this data - Discarded.
5 - Behavioural Features	Added weekend, weekday and private-label ratios as features.	8	1.4%	Cleanest structure so far; surfaced Weekday Convenience and Private-Label Loyalist - Best result so far.
7b - Threshold Tuning	Kept Sprint 5 features; reduced HDBSCAN min_samples from 50 to 20.	8	1.0%	Lowest noise to date; preserved the 8-cluster structure and revealed the Beer & FMCG buyer segment.

Sprint	Change introduced	Clusters	Noise	Outcome / status
9 - UMAP vs VAE + Store Sub-clustering	Ran the UMAP/VAE comparison and added store_id sub-clustering.	10	0.8%	Final outcome: joint UMAP compression won; 18 store-aware micro-segments; lowest noise of all trials - Final.

Note: the mid-project review presented Sprint 7b (8 clusters, 1.0% noise) as the interim final result. Sprint 9 superseded it by adding the UMAP-versus-VAE comparison and store-aware sub-clustering, yielding the final 10-cluster segmentation at 0.8% noise with 18 actionable micro-segments (24 leaf segments in total, including the six tribes that were not sub-divided). The 26-cluster figures shown for the Sprint 3 and Sprint 4 AE trials are the over-segmented results reported at the mid-project review; when the autoencoder was re-run on the final Sprint 9 data it under-segmented instead, producing 6 clusters at 1.7% noise (behavioural features appended) or only 2 clusters at 59.4% noise (behavioural features compressed jointly) - in all cases well short of the UMAP result.

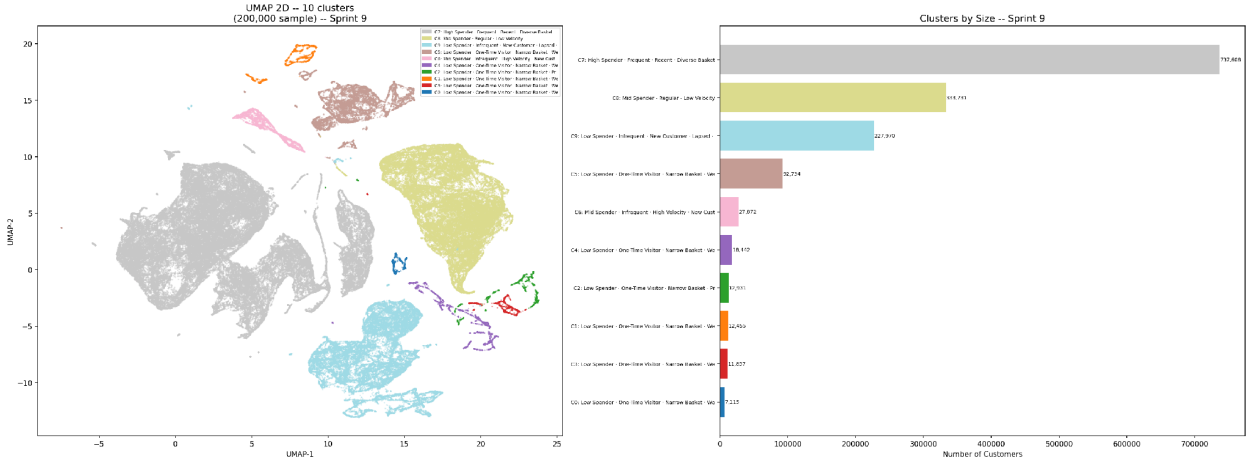


Figure E.2. Sprint 9 - final result: 10 tribes at 0.8% noise from joint UMAP + HDBSCAN, the lowest noise of all trials

Annex D - Segment Action Playbook (all 24 segments)

Tactical action plan for each of the 24 segments (the six standalone tribes plus the 18 sub-segments). For each segment it gives the profile, the recommended action, and what to avoid. This consolidates the playbook referenced in Section 6.4.

Segment (size)	Profile	Recommended action	Avoid / note
C7.0 Ultra-VIP (3,879)	€1,055, 26 visits, likely large families, 73% Store 7	'Carrefour Pro' tier: dedicated contact, advance promo notice, guaranteed Mahou 50cl and Red Bull stock at Store 7.	Do not discount; they are volume buyers, not deal seekers.
C7.1 VIP multi-store (50,321)	€659, 25 visits, split across Stores 7, 2 and 6	Cross-store app loyalty points; standardise top-5 product depth across stores.	Do not force single-store consolidation; the split is geographic.
C7.2 VIP very frequent (229,836)	€515, 17 visits	Fresh-counter quality SLA as a retention KPI; private-label tuna volume mechanic.	Operational action, not a CRM campaign.
C7.3 High spender frequent (453,572)	€418, 11 visits, 11-day recency (widest in C7)	Day-8 app push with a fresh offer before the 14-day drift threshold.	Do not invest in store conversion (81% Store 7, geographic).
C8.0 Frequent (17,275)	Mid-spend, frequent, recent	Treat as a C7 pipeline: category-expansion nudge and a 'Next Level' milestone reward.	Do not apply standard C8 mid-tier campaigns.
C8.1 Regular (102,779)	Mid-spend, regular, low velocity	Day-10 CRM alert with an FMCG or fresh staple offer; monitor C8 to C9 migration monthly.	Avoid broad discounts; align to known categories.
C8.2 Weekend-concentrated (213,677)	Highest churn risk in C8, Store 7 dominant	Friday PM push with a fresh or Mahou offer on the peak day.	Do not prioritise Store 6; Store 7 is dominant.
C9.4 One-time / lapsed (197,934)	€40, 1 visit, 1-day tenure, 59-day recency	Passive only: include in existing promo pushes at zero marginal cost.	No dedicated win-back budget; filter by app-installed first.
C9.3 Infrequent regular (23,271)	€71, 4 visits, 86-day tenure	Day-18 frequency nudge; olive oil plus beer bundle at a combined price.	Goal: lift from 4 to 6 visits per year.
C9.1 High-velocity promo (3,471)	57% promo, 36% PL, beer multipacks	Private-label beer substitution; make promos visible at entrance.	Do not expand category; the 2-product basket is structural.
C9.2 Regular beer buyer (1,445)	€98, 6 visits, Mahou 50cl and Heineken	Beer-volume stamp card; Store 6 stock audit; premium upsell.	Starkest store gap; fix Store 6 bulk-beer stock.
C9.0 Short-tenure (1,849)	€66, 2 visits, 11-day tenure	Day-10 welcome-back offer; in-store app-download incentive.	Critical first-30-day onboarding window.
C5.5 Weekend fresh mass (56,284)	€25, 1 visit, 100% weekend	Weekend fresh entrance display; strawberry, baguette and banana bundle.	Do not attempt weekday reactivation; 0% weekday is structural.
C5.4 Weekend 2-visit (13,581)	€47, 2 visits, 100% weekend	Day-18 nudge to pull a third visit; receipt QR to a weekend campaign.	Conversion pipeline to C5.3.
C5.3 Weekend regular (9,352)	€114, 3 visits, highest C5 value	Visit-3 app loyalty reward; tuna plus Mahou weekend bundle.	Highest recoverable C5 sub-segment; investigate Store 6 gap.
C5.2 PL loyalist (5,628)	89% private label, weekend	Private-label meal-deal bundle; ensure full PL range at Store 6.	Brand-loyal; they simply need the SKUs available.
C5.0 Weekend FMCG (4,070)	€60, 3 visits, multi-store	Fresh plus FMCG weekend bundle; monthly staples reminder.	Cross-store loyalty mechanic fits this segment.

Segment (size)	Profile	Recommended action	Avoid / note
C5.1 Electronics deal hunter (3,819)	95% promo, TVs and Switch, 87% Store 7	Friday electronics app push; accessory offer within 48 hours.	Do not expand to FMCG; do not replicate at small-format stores.
C6 Event shopper (27,872)	€99, 2 visits, 100% weekend, highest velocity, Clothing lift above 1.0	Second-visit offer 10 to 14 days after first; weekend family-pack bundle.	Maximise the single visit, not frequency.
C4 Weekday convenience (18,442)	€8, 1 visit, 1 product, 0% promo and PL	Point-of-sale impulse placement near the cold-drinks section.	No CRM or app spend; transient convenience visitors.
C2 PL loyalist weekday (12,931)	€5, 92% private label, bread	Private-label bakery stamp card; adjacent PL placement (butter, jam).	No discounting needed; expand basket within PL.
C1 Weekend impulse (12,455)	€11, 1 visit, weekend	Weekend impulse bundle at entrance (strawberry plus Coca-Cola).	In-store placement only; minimal loyalty signal.
C3 Weekday beer promo hunter (11,857)	€24, 41% promo, beer multipacks	Private-label beer shelf comparison with a clear price differential.	Do not run Mahou promos without a visible PL alternative.
C0 Extreme promo cherry-picker (7,115)	€17, 100% promo, 1 product	Passive monitoring; redesign the promotion structure (Decision 3).	No dedicated win-back budget.

All actions in this playbook come from observational segmentation: the lifts, gaps, and drifts show where to act, but do not themselves prove causal impact. Each action is a hypothesis to be validated before scaling, through one of two designs depending on the level of intervention.

Customer-level interventions (app-push messages, CRM offers, loyalty points) should run as clean randomised trials: within each segment, customers are randomly assigned to treatment and control, and success is measured as incremental visit frequency and revenue versus control, not only response rate. The C7.3 day-8 message should be judged on incremental visits over the next 30 days and the C8.1 day-10 message on the decline in C8-to-C9 migration, read off the quarterly migration matrix. A permanent universal control (2-5% per segment) prevents measurement bias from overlapping campaigns; small segments (C7.0, C9.2) lack power within a single cycle and should pool results across cycles.

Store-level interventions (the Store 6 SKU audit, the fresh-counter SLA, entrance displays) cannot be randomised with four stores and require quasi-experiments: difference-in-differences against unaffected stores, with customer composition held fixed by the segmentation, plus multi-format customers (the C7.1 tier) as a within-customer control.

Where neither design works, the expected effect size and metric must be registered before implementation and tested against a matched control drawn from other segments. Before-after uplift is never proof of effectiveness - it is confounded by seasonal and promotional calendars.

Annex E - Business Application with Streamlit

The Streamlit application is composed of 8 views, enabling the user to deeply review the insights enabled by our segmentation approach. The application has eight pages, reached from a left-hand navigation sidebar. An analyst typically starts on the Executive Overview for the headline numbers, opens the Cluster Deep Dive to understand a single segment, uses Cluster Comparison and Hidden Relationships to contrast segments and surface co-purchase patterns, and then moves to the Campaign Builder to size a treatment before launch. The screenshots below are representative views and are not exhaustive of the full application.

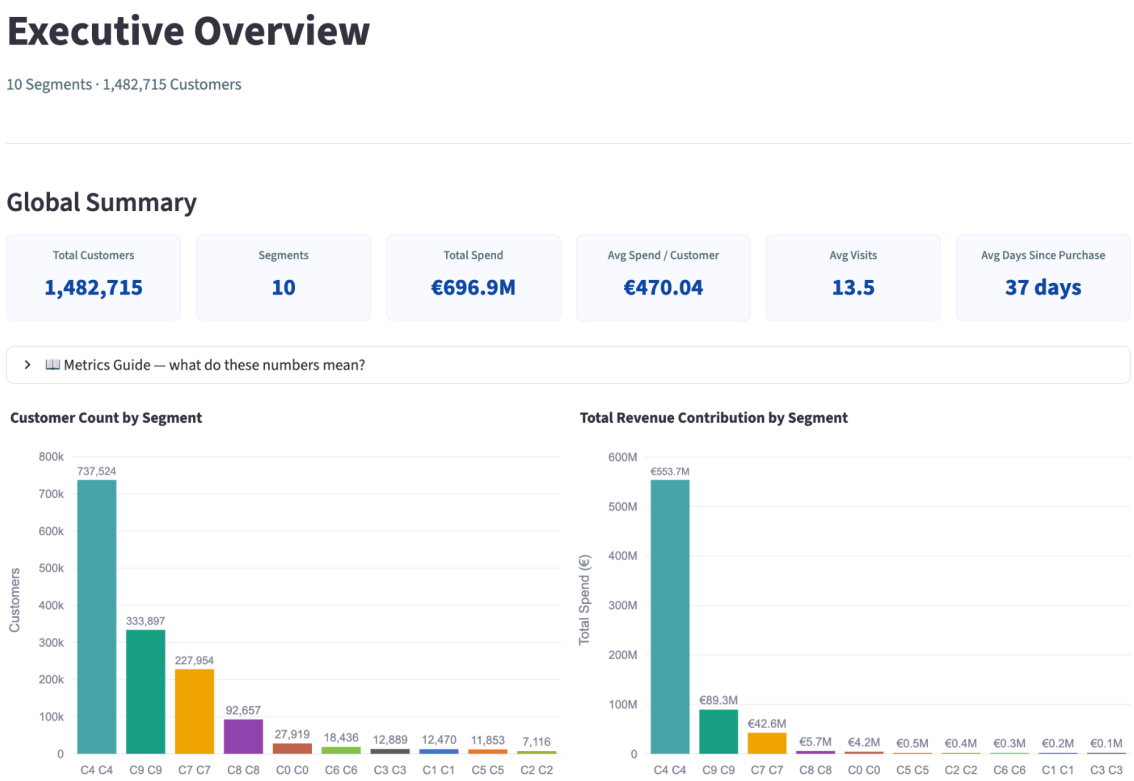
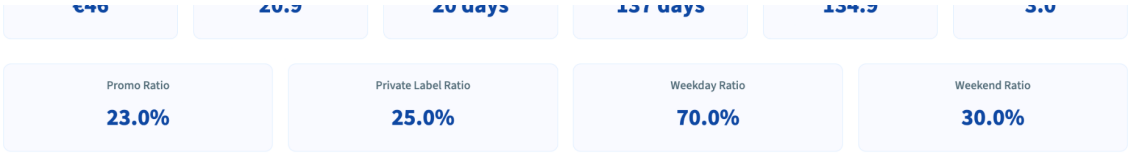
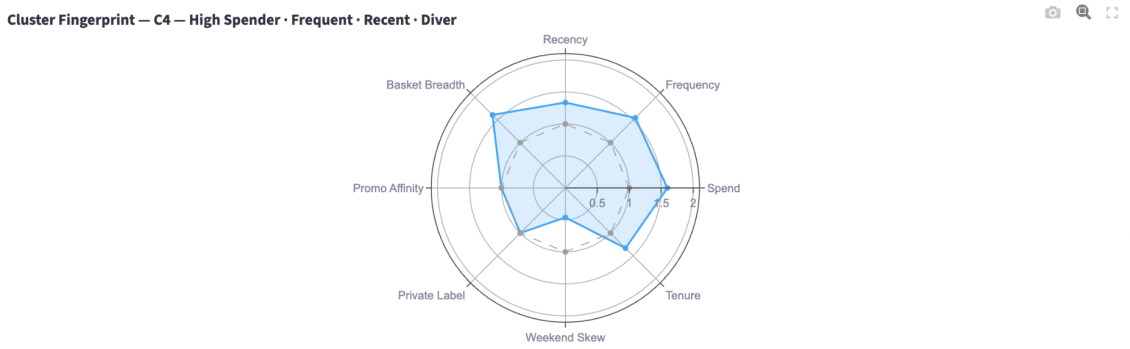


Figure G.1 - Executive Overview: global KPIs, per-segment customer counts and revenue contribution. Note: The picture displays a previous sprint, Sprint 5 values - but is today functionally equivalent to Sprint 9



Cluster Fingerprint

Normalised to global average = 1.0. Capped at 2.0.



Distributions vs Global Sample

Figure G.2 - Cluster Deep Dive: one segment's profile, behavioural metrics and signature products.

Red < 0.7x · White ≈ 1.0x (global average) · Green > 1.3x



Top Hidden Relationships

Sorted by lift descending.

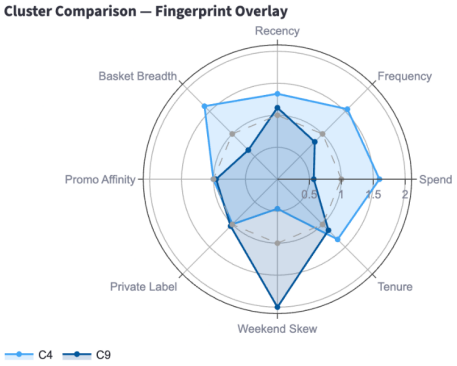
	Sector A	Sector B	Lift	Cluster	Co-Buyers	In-Cluster Rate	Global Rate
1	Electronics & Photo	Clothing	1.68x	C4 — High Spender · Frequent · Rece	190,688	25.9%	15.4%
2	Home & Bazaar	Electronics & Photo	1.58x	C4 — High Spender · Frequent · Rece	235,197	31.9%	20.2%
3	Electronics & Photo	Fresh Counter	1.57x	C4 — High Spender · Frequent · Rece	233,135	31.6%	20.1%
4	Electronics & Photo	FMCG	1.54x	C4 — High Spender · Frequent · Rece	237,966	32.3%	21.0%
5	Home & Bazaar	Clothing	1.51x	C4 — High Spender · Frequent · Rece	445,060	60.3%	40.1%

Figure G.3 - Hidden Relationships: co-purchase lifts and product associations within a segment.

Side-by-Side Metrics

Metric	C4	C9	Diff %	Interpretation
Customers	737,524	333,897	120.9	Higher
Total Revenue (€)	€553.7M	€89.3M	520.1	Higher
Avg Spend (€)	€751	€267	180.7	Higher
Median Spend (€)	€465	€162	187.8	Higher
Avg Visits	€21	€11	87.5	Higher
Median Visits	€13	€7	85.7	Higher
Avg Basket Size (€)	€46	€30	54.8	Higher
Days Since Last Purchase	€20	€30	-32.6	Better
Tenure (days)	€137	€116	18	Higher
Unique Products	€135	€54	150.3	Higher

Fingerprint Comparison



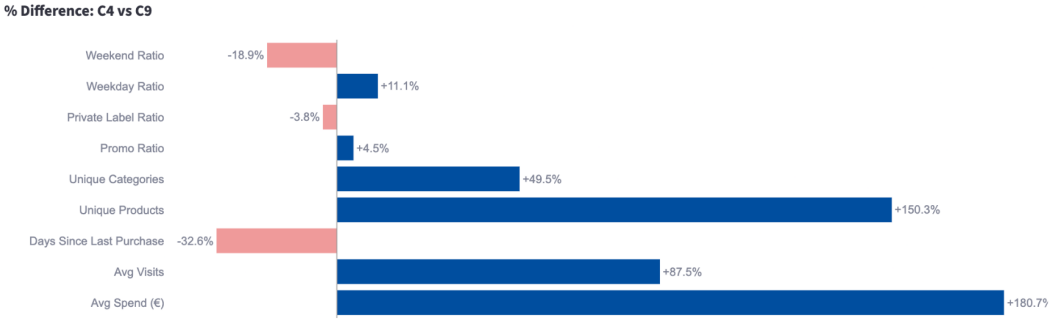


Figure G.4 - Cluster Comparison: side-by-side metrics and a fingerprint overlay for any two segments.

Customer Explorer

Individual Lookup | Customer Browser

Filter customers across multiple criteria and browse matching profiles.

Segments: C3 — Low Spend… ×

Top category: (all categories)

Min total spend (€): 0,00

Min visits: 1

Max total spend (€): 999999,00

Max days since last purchase: 365

12,889 customers match the current filters.

Rows per page: 50

Showing rows 1–50 of 12889

customer_id	cluster_label	total_spend	avg_basket_size	total_visits	days_since_last_purchase	promo_ratio	private_label_ratio	top_category
42404141a85f6e08acae8bdeb1135a62017efd14a6dacc5d3485d9973a9b3e0d	C3	24.11	4.82	5	23	0	1	Food & Grocery
426915450b50def8fd8b4d712b8c978ccec2c5c0402a609b938807b94c25ae22	C3	3.15	3.15	1	6	0	1	Food & Grocery
42d1d39e86488e3f2433f7a40bc1e23e3ee0b62deeb799f8160ec8ba8402fb34	C3	2.54	2.54	1	73	0	1	Food & Grocery
3c41c837c90716d98c7d63abcbfb41200577a97db17a2f7b03db58706a8b76e6	C3	1.3	1.3	1	69	0	1	Food & Grocery
43212a6871870dbcd539d49c53710d479e209f0e7407a2138b37b61a7cd73fee	C3	2.66	2.66	1	16	0.3333	0.6541	Food & Grocery

Figure G.5 - Customer Browser: drill down to individual customers and their attributes.

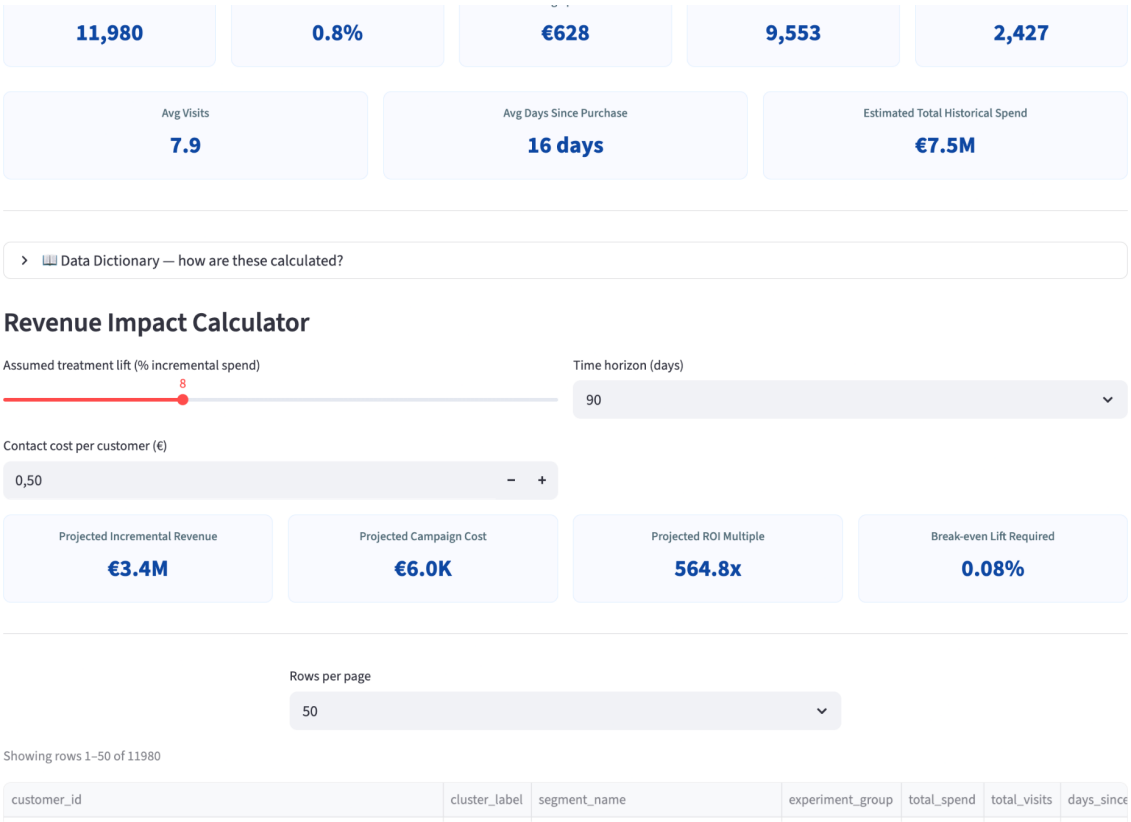


Figure G.6 - Campaign Builder: a revenue-impact calculator to size a treatment before launch.

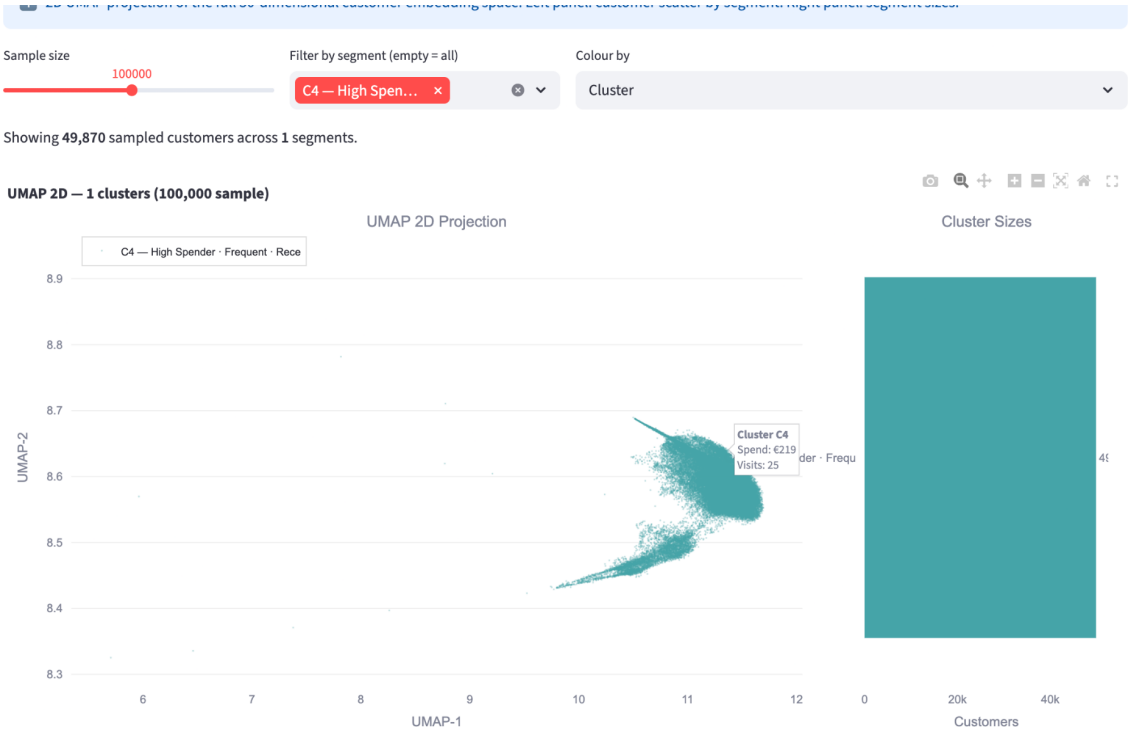


Figure G.7 - Visual Explorer: an interactive 2D UMAP map of the base coloured by segment.

Data Quality & Methodology

Loaded Files

File	Status	Rows	Columns	Size (MB)	Note
Customer Vectors (customer_vectors_sprint9.parquet)	✅ Loaded	1,482,715	21	15096.6	
Cluster Assignments (UMAP) (customer_clusters_sprint9_UMAP.parquet)	✅ Loaded	1,482,715	52	399.8	
Cluster Behavioral Profile (cluster_profile_behavioral_sprint9.csv)	✅ Loaded	10	36	0	
Cluster Rich Profile (cluster_profile_rich_sprint9.csv)	✅ Loaded	50	9	0	
Cluster Summary (cluster_summary_sprint9.csv)	✅ Loaded	10	6	0	
Co-Purchase Data (co_purchase_sprint9.csv)	✅ Loaded	58	11	0	
Product Co-Purchase (product_copurchase_sprint9.csv)	✅ Loaded	29	12	0	

Join Quality

Vectors rows	Cluster rows	Joined rows	Unmatched (vectors)	Unmatched (clusters)
1,482,715	1,482,715	1,482,715	0	0

Cluster Health

Number of clusters	Smallest cluster	Largest cluster	Imbalance ratio
10	7,116	737,524	103.6x

Figure G.8 - Data Quality & Methodology: coverage checks, data lineage and the methodology summary.

Annex F - Production Deployment

The pipeline was created and validated using a single, deliberately constrained workstation set-up. It was enough to process 1.48 million customer cases through the entire pipeline, although a number of the software tools involved were selected for their suitability. However, the deployment at full-scale - over 11 million El Club Carrefour loyalty programme participants on a longer timespan - will need a more sustainable, cloud-based architecture.

For production, the same logical pipeline maps onto a managed cloud stack. With Apache Spark re-write, a managed Databricks Lakehouse environment would be recommended. It consolidates the distributed Spark layer, the GPU compute, the governed storage, and the model lifecycle into one platform. Databricks is cloud-agnostic and the data can be localised in the EU, which is especially relevant given the volume of the customer data.

Pipeline stage	Local implementation	Production Environment
Raw data & checkpoints	On-disk Parquet on a single workstation, accessed directly by DuckDB.	Delta Lake tables in cloud object storage, governed by Unity Catalog, with provenance, lineage and access control + EU residence.
Data preparation & feature engineering	DuckDB out-of-core SQL (up to 8 GB, 8 threads, disk spill) on 191M rows.	Spark SQL and DataFrame operations on a cluster of managed nodes (autoscaling), aggregation done across all historical multi-billion-row dataset.
Product text embedding	OpenAI text-embedding-3-small (1,536-dimensional vector) through external API call.	Same OpenAI API accessed from a Databricks job in batches over the de-duplicated catalogue; costs generated only when the catalogue is updated, not for each individual customer.
Item2Vec training	gensim locally on CPU; some experiments on a cloud GPU.	gensim training job on standard clusters scheduled along with embedding and reduction steps.
Dimensionality reduction (UMAP)	Umap-learn CPU (low_memory = True); fit on a 150k subsample, transformed in batches.	RAPIDS cuML on GPU clusters; fit on the entire customer dataset using multiple GPUs if necessary.
Clustering (HDBSCAN)	hdbscan on CPU	GPU-accelerated cuML HDBSCAN on the same GPU clusters.
Artefact persistence	Parquet and pickled models on the local disk.	MLflow tracking combined with Unity Catalog model registry, with product embeddings and customer matrix as version-controlled Delta tables.
Business serving (Streamlit)	Executed locally for inspection only.	Databricks Application with business logic deployed on Delta Lake tables; business interface sits side-by-side with data.

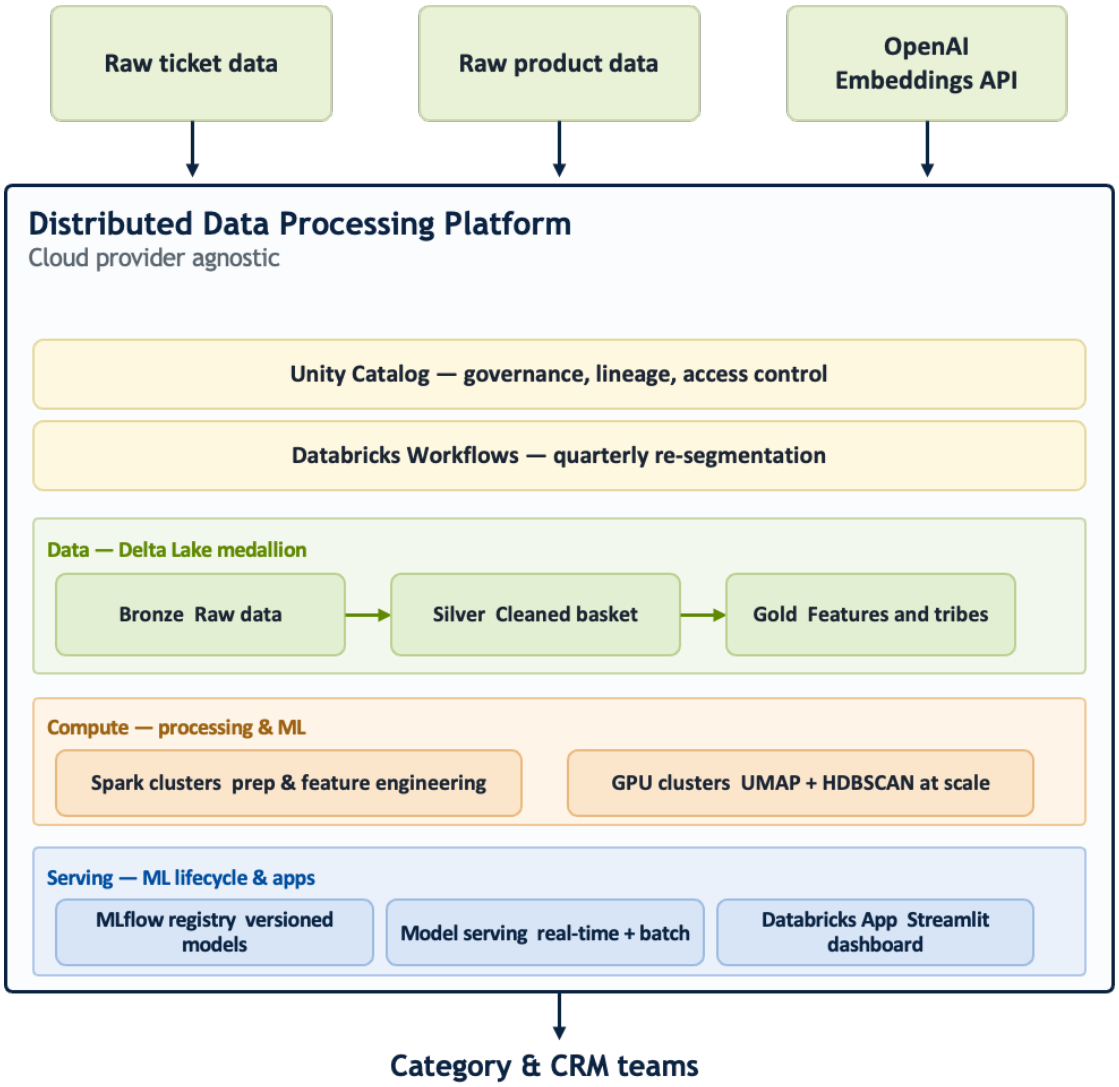


Figure H.1 - Production architecture

Development vs production environment

The development build ran on a single laptop and relied on three workarounds to fit within 8 GB of RAM: out-of-core SQL in DuckDB with disk spill, UMAP fitted on a 150,000-customer sample, and checkpointed Parquet caching so each stage could resume. A production environment removes these constraints by providing distributed processing (Spark) and GPUs (RAPIDS cuML), so UMAP and HDBSCAN run on the full customer base rather than a sample, and the segmentation is reproducible at the full eleven-million-customer scale. Figure H.2 maps the pipeline onto that production environment and notes the development-side workaround beneath each stage.

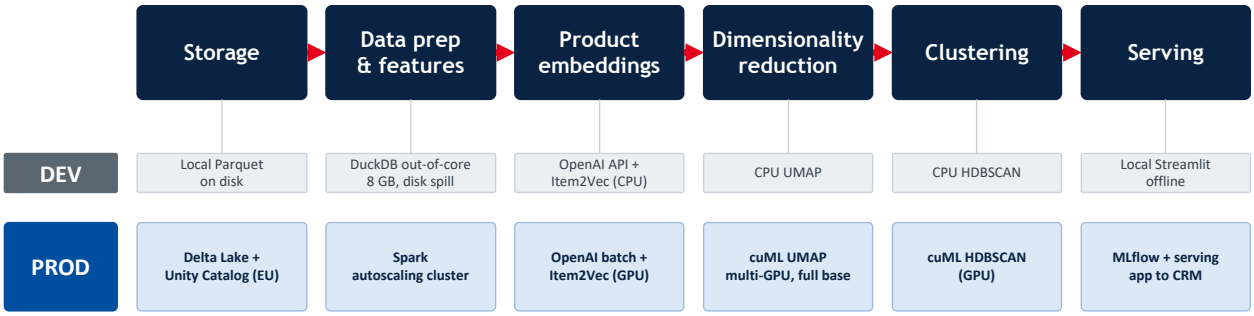


Figure H.2 - Production architecture (distributed processing and GPUs), with the development-environment workaround noted under each stage.

Pipeline run cadence

Not every stage needs to run at the same frequency. The product space is comparatively stable, customer behaviour drifts month to month, and new customers arrive continuously, so the pipeline is run on a tiered cadence, summarised in the matrix below.

Cv	Real-time	Monthly	Quarterly	On demand	Remarks
New-customer tribe scoring	✓				Scored on first transaction; no retraining.
Customer vectors & features		✓			Recomputed from the latest transactions; dashboards refreshed.
Full re-segmentation (UMAP + HDBSCAN)			✓		Re-derive and re-name tribes; capture accumulated drift.
Product embeddings (OpenAI)				✓	Only when the product catalogue changes materially.
Dashboards / Streamlit		✓	✓		Metrics monthly; full refresh each quarter.

In short: the costly stages stay infrequent, the customer view stays fresh, new customers are assigned immediately, and a full, reproducible re-segmentation lands every quarter.